Implementation Report

Inventory management system- GitHub Repository

The Grocery Shop Inventory Management System was implemented using C11 and C++17. It provides a command-line interface for managing products, suppliers and stock transactions. The system supports adding, editing and removing records, searching and sorting products, recording stock movements, and generating current-stock, low-stock and total-inventory-value reports.

The application uses structured in-memory data and stores records in a text file divided into the sections [PRODUCTS], [SUPPLIERS] and [TRANSACTIONS]. This enables inventory data to be saved and reloaded when the application restarts.Testing covered invalid menu input, negative product quantities, duplicate product names, low-stock reporting, inventory value calculation and persistence across restarts. Invalid quantities and duplicate names were rejected. The low-stock report correctly identified Sugar with 7 units against a threshold of 8, while the total inventory value was reported as 234.01. During missing-file and malformed-file testing, the application displayed an empty stock report without crashing. Further verification is needed to confirm complete malformed-record handling.

Future improvements include expanding the sample dataset with more realistic products and suppliers, strengthening validation and recovery for malformed inventory files, and introducing automated tests for empty inventories and invalid records. Additional enhancements could include more advanced searching, sorting and reporting features.